DICE Progressive & Milestone Billing: Product Requirements Document

Version: Draft v1.2 (expanded from v1.1) | **Date:** 24 October 2026 | **Audience:** Product, Engineering, QA, Procurement, AP, Finance

> **Core design principle:** Keep the existing PO → GRN/SRN → Invoice → Matching → Approval → AP workflow. Add a PO billing configuration and a shared entitlement/allocation layer. Add milestone scheduling and approval as net-new capability. Add certified work-progress claims only where the contract permits billing before accepted receipt evidence exists.

1. Executive Summary

DICE today supports one billing pattern: a supplier invoices against a PO and its accepted GRN/SRN. Many real contracts do not fit this pattern. Construction, installation, IT implementation, AMC and project-based services are billed either **in instalments as work progresses** (progressive billing) or **on completion of contractual deliverables** (milestone billing).

This PRD defines what DICE must add so that both methods run through the *existing* invoice submission, matching, approval and AP processes. We do not build a second invoice engine. We add:

1. A **Billing Configuration** on the PO (method, basis, acceptance gates).
2. A shared **Billing Entitlement & Allocation layer** that answers "how much can this supplier invoice now, and how much has already been consumed?"
3. A **Milestone module** (schedule, evidence, approval) as net-new capability.
4. An optional **Progress Claim module** for certified work-progress billing.

Release recommendation: Phase 0 fit-gap, then the shared allocation layer and receipt-based progressive billing, then milestone billing, then (only if the business confirms the need) certified work progress.

2. Problem Statement

- Suppliers on long contracts cannot be paid in stages without workarounds (splitting POs, manual overrides, off-system tracking).
- There is no contractual schedule in DICE, so buyers cannot enforce "pay only when deliverable X is accepted".

- Where partial invoicing exists, consumption of accepted quantity may be tracked loosely, creating risk of double billing and PO overrun.
- AP and Procurement lack a single view of what is contracted, accepted, invoiced, paid and remaining.

3. Goals, Non-Goals and Success Metrics

3.1 Goals

- G1: Allow a PO to be configured as Standard, Progressive or Milestone with no change to existing POs.
- G2: Guarantee that no quantity or value is invoiced twice or beyond entitlement, even under concurrency.
- G3: Give buyers contractual control: invoices become possible only after the configured acceptance gate passes.
- G4: Reuse existing matching, tax, approval and AP flows with no parallel engine.
- G5: Provide full auditability and a clear balance view (contracted / eligible / reserved / invoiced / paid / remaining).

3.2 Non-Goals (v1)

- Replacing the PO amendment workflow, GRN/SRN workflow or AP posting.
- Automatic payment scheduling, retention release automation and advance recovery logic (flagged as decisions in Section 28).
- Supplier-entered progress percentages creating eligibility without buyer certification.
- Multi-currency re-valuation of schedules.

3.3 Success Metrics (to be baselined in Phase 0)

Metric	Target
Overbilling or duplicate allocation incidents in production	0
Standard-PO regression defects	0 critical, 0 high
Invoices on Progressive/Milestone POs processed without manual workaround	≥ 95%
Median time from milestone submission to approval	Tracked; target agreed with Procurement

Metric	Target
Invoice rejections caused by "wrong reference / ineligible allocation"	Decreasing release over release

4. Users and Roles

Role	Needs
Buyer / PO creator	Configure billing method, define milestone schedule, set acceptance rules.
Project / service owner	Review milestone evidence, accept or reject, certify work progress.
Warehouse / service receiver	Raise and accept GRN/SRN (unchanged process).
Supplier	See what is billable, submit completion evidence, create invoices against eligible items.
AP / Finance	See billing method and allocations on invoices; handle reversals and credit notes.
Procurement manager	Approve schedules and amendments; monitor exposure and exceptions.
Auditor / admin	Trace every schedule change, approval and allocation event.

5. Glossary

Term	Meaning
Entitlement	Net quantity/value a supplier is currently allowed to invoice.
Allocation	Link from an invoice line to an entitlement source (receipt line, claim or milestone) with an amount.
Reservation	Temporary allocation held while an invoice is in flight (not yet posted).
Commit	Allocation that is final once the invoice passes the configured status.

Term	Meaning
Billing base	The net-of-tax value (PO total or selected lines) against which schedule percentages apply.
Acceptance gate	A required approval before something becomes billable (GRN/SRN, milestone approval, progress certification, or combination).
Progress claim	A record of cumulative measured work for a period, certified by the buyer.

6. Product Decision: What Each Billing Type Means in DICE

Dimension	Progressive billing	Milestone billing
Invoice eligibility trigger	Accepted receipt quantity/value (GRN/SRN), or separately certified work progress	Completion and approval of a predefined contractual deliverable
Where progress is recorded	Existing GRN/SRN (receipt mode); new Progress Claim (work-progress mode)	New PO Milestone Schedule and Milestone Completion/Approval record
Supplier selects on invoice	PO + eligible GRN/SRN lines, or a certified claim	PO + approved milestone ID (+ SRN reference where required)
Typical use	Phased deliveries, framework call-offs, long-running supply and services, measured project works	Fixed-price projects, implementations, installations, design-build, commissioning
System must prevent	Reusing accepted quantity/value; billing beyond remaining entitlement	Invoicing unapproved milestones; duplicate allocations; amounts above milestone balance
Reuse	Existing PO, GRN/SRN, invoice, matching, tax, approval, AP	Same, plus SRN where applicable
New	Billing flag, allocation/balance controls; optional Progress Claim	Schedule builder, status/approval, evidence, allocation, version history

7. Key Design Principles

1. **Three separate concepts.** The PO defines the commercial agreement and billing method. The GRN/SRN records receipt/acceptance. The Billing Entitlement layer decides what is invoiceable and what is consumed.
2. **A GRN is not a milestone**, and a milestone approval is not automatically an SRN. A milestone may require an SRN as an additional gate. Do not create duplicate receipt records to represent milestone completion.
3. **Standard flow is untouched.** Existing POs and the default Standard method behave exactly as today.
4. **Allocation is a record, not a flag.** Never use a single "invoiced" boolean as the control.
5. **Net, tax and gross stay separate.** Entitlement is a net amount; apply existing tax rules. Paid ≠ invoiced ≠ allocated.
6. **Supplier cannot change the method.** Invoices inherit the PO's billing method.
7. **Configuration over code.** Acceptance gates are configurable by PO category.

8. Existing and Proposed Flows

Existing standard flow: PO created and approved → goods received → GRN submitted/accepted → supplier submits invoice against PO/GRN → existing matching and validations → invoice approval → AP.

Receipt-based progressive flow: PO (Progressive, basis Accepted Receipt) → GRN/SRN via existing process → accepted quantity/value becomes eligible → supplier selects receipt lines → DICE checks unallocated eligible quantity/value and PO limit → reservation → matching → approval/AP.

Certified work-progress flow: PO (Progressive, basis Certified Work Progress) → Progress Claim submitted for period/cumulative completion → buyer certifies measured value → DICE calculates newly certified, unconsumed amount → required SRN/acceptance gate checked → supplier invoices against claim → matching → approval/AP. Needed only when the contract allows billing measured work before it is represented by an accepted GRN/SRN.

Milestone flow: PO with milestone schedule → schedule validated and approved with PO → supplier/project user submits completion evidence → authorised approver accepts → required SRN accepted where applicable → milestone becomes invoice-eligible → supplier invoices PO + milestone → DICE checks remaining milestone entitlement → matching → approval/AP.

9. Progressive Billing in Depth

9.1 What it is

Progressive billing lets a supplier invoice repeatedly against one PO as work or deliveries are accepted, with each invoice limited to the *cumulative accepted but not yet invoiced* amount. Typical triggers: staged deliveries, monthly service acceptance, or measured percentage completion.

9.2 Two bases

Basis	Evidence of progress	When to use	Release
Accepted GRN/SRN (default)	Accepted receipt lines	Goods in phased deliveries; services accepted via SRN	Release 1
Certified Work Progress	Buyer-certified progress claim	Measured works where billing precedes receipt evidence	Later release, only if confirmed

9.3 Receipt-based worked example

PO line: 100 units × AED 1,000 = AED 100,000 net.

Event	Quantity	Value (AED)
GRN-001 accepted	30	30,000
GRN-002 accepted	40	40,000
GRN-003: 10 received, 7 accepted, 3 rejected	7	7,000
Total eligible	**77**	**77,000**
Invoice 1: 30 units from GRN-001	30	30,000
Invoice 2: 20 units from GRN-002	20	20,000
Remaining eligible	**27**	**27,000**
Not yet eligible (not accepted)	23	23,000

Rejected units are excluded permanently. Pending units become eligible only on acceptance.

9.4 Certified work-progress worked example

PO value AED 100,000. Previously certified cumulative: AED 30,000. New cumulative certified: AED 50,000. **Current claim entitlement = AED 20,000**, not AED 50,000. It becomes invoiceable only after certification and all configured gates.

9.5 How it is implemented

1. **PO configuration:** add `billing_method`, `progressive_basis`, `partial_billing_allowed`, `acceptance_gate`.
2. **Eligibility calculation:** accepted quantity/value of each receipt line, minus active reservations and committed allocations, bounded by the PO remaining limit.
3. **Invoice UI:** supplier selects PO, then eligible receipt lines; the screen shows *accepted, previously allocated, currently allocating, remaining* per line.
4. **Validation:** reject overbilling, ineligible status, duplicate allocation and PO overrun, with an actionable message.
5. **Reservation and commit:** reserve on submission inside the transaction; commit per invoice status policy; release on rejection/cancellation.
6. **Reversals:** posted invoices use existing credit note/reversal processes. Allocations are never silently deleted.
7. **Receipt corrections:** a reduced GRN/SRN reduces *future* entitlement; posted invoices are preserved and flagged for review if entitlement falls below consumption.

9.6 Progress Claim lifecycle (certified mode)

`Draft → Submitted → Under Review → Certified | Rejected | Returned for Info → (Partially/Fully) Invoiced → Closed`

- A claim stores period, line scope, measurement basis, cumulative claimed and certified value/percentage, current claim value, evidence, submitter, approver, timestamps and version.
- Certified cumulative value may never fall below previously consumed value. Corrections use a new version or an authorised adjustment claim.
- A supplier-entered percentage never creates eligibility on its own.

10. Milestone Billing in Depth

10.1 What it is

Milestone billing ties payment to **contractual deliverables**. The schedule is part of the commercial agreement, approved with the PO, and each milestone becomes billable only after its configured acceptance gates pass.

10.2 Schedule design options

Option	Description	Validation
Fixed amount	Each milestone has an absolute net amount	Sum equals billing base
Percentage	Each milestone is a % of the billing base	Sum equals 100%
Line-scoped	Milestones tied to specific PO lines	Sum per line equals that line's value
PO-scoped	Milestones tied to the whole PO	Sum equals PO net total

Do not create duplicate PO lines just to represent milestones. Use separate lines only if scope, quantity, accounting or tax treatment differs.

10.3 Worked example

Milestone	Amount (AED)	Trigger	Initial state
M1 Design Approval	20,000	Design deliverable accepted	Pending
M2 Installation Complete	50,000	Installation accepted	Pending
M3 Commissioning Complete	30,000	Commissioning accepted	Pending
Total	**100,000**	100% of PO net billing base	Must validate

M1 journey: Pending → supplier submits evidence → buyer approves → required SRN accepted (if applicable) → **Eligible AED 20,000** → invoice allocates AED 20,000 → **Fully Invoiced, remaining AED 0**. If M1 is approved but the required SRN is pending, the invoiceable amount stays AED 0. A second full-value invoice on M1 is blocked.

10.4 Milestone lifecycle (two independent dimensions)

Avoid one overloaded status. Track two:

- **Acceptance status:** Draft → Pending Completion → Submitted → Approved | Rejected | Info Requested → Cancelled.
- **Billing status (derived):** Not Eligible → Eligible → Partially Invoiced → Fully Invoiced. Computed from acceptance, gates and allocations.

10.5 Gating rules

A milestone is **Eligible** only when all are true: the schedule version is approved; acceptance status is Approved; every configured gate (e.g. SRN accepted) is satisfied; and the remaining balance is above zero.

10.6 How it is implemented

1. **Schedule builder** on the PO: add/edit/reorder milestones with name, amount or %, scope, planned date, acceptance criteria, evidence requirements, approver and optional SRN requirement.
2. **Validation** at save and again at PO approval: totals reconcile to the billing base; no negative or zero amounts; unique milestone IDs; planned dates in logical order (warning, not block).
3. **Finalisation:** PO approval locks the schedule as Version 1.
4. **Completion workflow:** supplier or project user submits evidence (date, remarks, attachments). An authorised approver accepts, rejects or requests more information; actor, timestamp and reason are recorded.
5. **SRN gate:** where the PO category requires it, DICE links the milestone to an SRN and holds eligibility until the SRN is accepted.
6. **Invoice creation:** supplier sees only eligible milestones with contractual, allocated and remaining amounts. Partial allocation is allowed only if enabled on the PO.
7. **Amendments:** changes after approval or first invoice go through the existing PO amendment path, produce a new schedule version, and preserve historic allocations.
8. **Cancellation:** a milestone with allocations cannot be cancelled; it must be amended with a documented reason.

10.7 Milestone variants to consider

Variant	Handling in v1
Advance payment milestone (e.g. 10% on PO acceptance)	Supported as a milestone with trigger "PO approved" and no evidence; confirm with Finance (advance rules)
Retention (e.g. 10% held until final acceptance)	Out of v1 unless Finance confirms the existing retention rule; schedule must not double count
Partial milestone invoicing	Controlled by *Partial Billing Allowed*; default is one invoice per milestone
Parallel milestones	Supported; each has independent state
Dependent milestones (M2 requires M1)	Optional dependency field; warning only in v1

11. PO Billing Configuration Requirements

Field	Behaviour
Billing Method	Standard (default), Progressive, Milestone
Progress Measurement Basis	Shown for Progressive: Accepted GRN/SRN (default) or Certified Work Progress
Billing Schedule	Shown for Milestone; at PO or PO-line scope
Schedule Billing Base	PO total or selected lines, net of tax unless rules state otherwise
Partial Billing Allowed	Allows partial allocation to a receipt line or milestone
Required Acceptance Gate	By PO category: GRN/SRN, milestone approval, progress certification, or combination
Schedule Version	System-controlled; history preserved
Approver / Evidence Rules	Required for milestone schedule and certified progress

PO validation rules

- Existing POs keep current behaviour and are not silently migrated.
- Billing method is inherited by invoices; suppliers cannot change it.
- A schedule must not exceed its billing base; percentage schedules total 100% before finalisation unless a formally supported exception exists.
- After first receipt/invoice or schedule approval, changes need existing amendment controls and approval.
- Method can be changed only on a PO with no receipts, invoices or allocations; otherwise it requires amendment and Procurement approval.

12. Billing Entitlement and Allocation Layer

Architecture: one shared allocation service with pluggable *entitlement providers*: Receipt provider (GRN/SRN lines), Claim provider (certified claims), Milestone provider (approved milestones).

`PO billing config → Entitlement provider → Eligibility calculation → Transactional reservation → Existing matching → Approval → AP`

12.1 Core calculations

- **Receipt eligible** = accepted receipt entitlement − active reservations − committed allocations (adjusted for receipt corrections).
- **Milestone eligible** = approved milestone amount (if all gates pass) − active reservations − committed allocations.
- **Claim eligible** = certified claim value not previously consumed, after gates.
- **PO remaining limit** is a separate guardrail checked in addition to every provider balance.
- Keep net, VAT and gross distinct; apply existing tax rules.

12.2 Allocation states

`Reserved → Committed → Released` (on rejection/cancellation before posting) or `Committed → Reversed` (via credit note/reversal after posting).

The point at which Reserved becomes Committed (invoice submitted, approved or posted) is an open decision (Section 28) and must follow the existing invoice status policy.

12.3 Concurrency and idempotency

- Revalidate entitlement inside the transaction that commits the reservation (row-level lock or optimistic version check on the entitlement balance).
- Two simultaneous invoices cannot consume the same remaining amount.

- Idempotency key per submission protects against retries and double-clicks.
- A background job releases stale reservations on abandoned drafts per policy.

13. Functional Requirements

ID	Requirement	Priority
FR-01	PO supports Billing Method with Standard as default; existing POs unchanged.	Must
FR-02	Progressive POs require a progress basis.	Must
FR-03	Receipt-based eligibility excludes pending and rejected quantities.	Must
FR-04	Invoice UI shows accepted, previously allocated, current and remaining per selectable line.	Must
FR-05	Allocation is blocked if above line, entitlement or PO remaining balance.	Must
FR-06	Reservation is transactional and idempotent.	Must
FR-07	Reservations are released on rejection/cancellation before posting.	Must
FR-08	Posted invoices are adjusted only via reversal/credit note.	Must
FR-09	Milestone schedule builder with fixed or percentage amounts and PO/line scope.	Must
FR-10	Schedule must reconcile to billing base before finalisation.	Must
FR-11	Milestone completion submission with evidence; approval/rejection with reason.	Must
FR-12	Milestone eligibility honours the configured SRN gate.	Must
FR-13	Invoice references PO + milestone; duplicate or excess allocation blocked.	Must
FR-14	Schedule amendments are versioned; history and allocations preserved.	Must

ID	Requirement	Priority
FR-15	Role-based permissions for configuration, submission and approval.	Must
FR-16	Full audit trail for schedule, approval, allocation, release and reversal events.	Must
FR-17	Notifications for submitted, approved, rejected and eligible milestones/claims.	Should
FR-18	Pending approval queue for milestones and claims.	Should
FR-19	PO balance view: contracted, eligible, reserved, invoiced, paid, remaining.	Should
FR-20	Progress Claim submission, certification and cumulative calculation.	Could (only if confirmed)
FR-21	Bulk import of milestone schedules (CSV/Excel) with validation.	Could
FR-22	Reports: open milestones, aged approvals, billing vs plan.	Should

14. Non-Functional Requirements

- **Integrity:** zero tolerance for double allocation; verified by concurrency tests.
- **Performance:** eligibility lookup and validation must not noticeably slow invoice creation; targets set in Phase 0 against current baselines.
- **Auditability:** immutable event log of who, when, what changed, before and after values.
- **Security:** role-based access; suppliers see only their own POs, milestones and claims; evidence files follow existing attachment security.
- **Backward compatibility:** all existing APIs, integrations and reports behave identically for Standard POs.
- **Observability:** metrics and alerts on allocation failures, stale reservations and balance mismatches.
- **Retention:** follow existing financial-record retention policies.
- **Currency:** AED and multi-currency handling follow current DICE rules.

15. Roles and Permissions (indicative)

Action	Buyer	Project owner	Supplier	AP	Procurement mgr
Configure billing method/schedule	Yes	No	No	No	Approve
Submit milestone evidence / progress claim	No	Yes	Yes	No	No
Approve milestone / certify progress	No	Yes	No	No	Escalation
Create invoice against entitlement	No	No	Yes	No	No
View allocations and balances	Yes	Yes	Own POs	Yes	Yes
Reverse allocation via credit note	No	No	No	Yes	No
Amend schedule	Request	No	No	No	Approve

Segregation of duties: whoever submits completion evidence cannot approve the same milestone. Final role mapping to be confirmed with Procurement.

16. Data Model

These are logical entities. Engineering must inspect existing DICE schemas and reuse equivalents before creating new tables.

Entity	New/Changed	Key fields
PO billing configuration	Changed	billing_method, progressive_basis, partial_allowed, acceptance_gate, schedule_version
PO milestone schedule	New	milestone_id, po_id, line_id, name, amount/percentage, planned_date, criteria, approver, version, acceptance_status
Milestone completion	New	milestone_id, actual_date, remarks, attachments, submitted_by/at, decided_by/at, decision, reason
Progress claim	New (optional)	claim_id, po_id, period, cumulative_claimed, cumulative_certified, current_claim, evidence, status, version

Entity	New/Changed	Key fields
Invoice allocation	Reuse or create	invoice_line_id, source_type (receipt/claim/milestone), source_id, net_amount, quantity, state (reserved/committed/released/reversed), idempotency_key
Audit event	Reuse or extend	entity, event_type, actor, timestamp, before, after

Relationships: PO 1—N GRN/SRN; PO 1—N Milestones; PO 1—N Progress Claims (optional); Invoice N—N (Receipt lines | Milestones | Claims) through the Allocation entity.

17. Service Responsibilities

Service	Responsibility
PO billing configuration	Persist method/basis, validate permissions and setup
Billing schedule service	Create/update milestones, validate totals, version, amendments
Receipt entitlement service	Read eligible accepted lines and remaining quantity/value
Milestone workflow service	Submit, approve/reject, link SRN, compute eligibility
Progress claim service (optional)	Submit, certify, compute cumulative entitlement
Billing allocation service	Reserve, commit, release and reverse transactionally
Invoice validation/matching	Call entitlement checks alongside existing PO, receipt, duplicate, tax and category checks
Audit/reporting service	History and balances

Illustrative operations: `getEligibleEntitlements(po)`, `reserveAllocation(invoice, items, idempotencyKey)`, `commitAllocation(invoice)`, `releaseAllocation(invoice)`, `reverseAllocation(creditNote)`. Names must align with DICE conventions. **Do not build a second matching engine.**

18. Screens to Add or Change

Screen	Change
PO Create/Edit	Billing Configuration; conditional basis or milestone schedule
PO Detail	Billing Summary and Milestones tabs with eligible, reserved, invoiced, approved, paid, remaining
GRN/SRN Detail	Show invoice allocations and remaining eligible quantity/value
Milestone List/Detail	Schedule, evidence, approval, SRN link, allocations, audit
Progress Claim List/Detail	New, only if certified mode is in scope
Supplier Invoice Create	Dynamic selector: receipt lines, claim or approved milestone
Invoice Matching	Entitlement checks, allocated amounts, remaining balance, actionable errors
Approval Queue	Pending milestones and claims alongside invoices
AP Register	Billing method and linked receipt/milestone reference

Error message examples: "Only AED 20,000 remains eligible on Milestone M2; you entered AED 25,000." / "Milestone M3 is approved but awaiting SRN acceptance and cannot be invoiced yet."

19. Notifications

- **Supplier:** milestone approved, rejected or info requested; milestone or receipt became eligible; invoice blocked by entitlement.
- **Approver:** milestone or claim awaiting review; reminder after configurable ageing.
- **Buyer / Procurement:** schedule amendment pending; PO nearing fully billed.
- **AP:** invoice reversal affecting an allocation.

20. Reporting

- **PO billing status:** contracted, eligible, invoiced, paid, remaining by PO, line and milestone.

- **Milestone pipeline:** planned vs actual dates, ageing of pending approvals.
- **Exceptions:** blocked invoices by reason, stale reservations, amended schedules.
- **Audit export** by PO.

21. Edge Cases and Business Rules

- **Partially accepted GRN:** only the accepted portion is eligible.
- **Receipt reversal / SRN reduction:** re-evaluate future entitlement; preserve posted history; flag cases where consumed exceeds entitlement.
- **Invoice rejected before posting:** release the reservation per status policy.
- **Invoice posted then adjusted:** use the credit note/reversal path.
- **Multiple GRNs on one invoice:** allocate per receipt line and validate the combined PO limit.
- **Partial invoice on the same GRN:** only within remaining quantity/value.
- **Milestone approved, SRN pending:** not eligible.
- **Milestone amount changed after invoices:** amendment and versioning; preserve historic allocations.
- **PO value amended:** future entitlement follows approved rules; historic certified amounts untouched.
- **Duplicate supplier invoice number:** the existing duplicate control applies.
- **Tax, retention, advance:** decide whether they affect net entitlement or payable only; never assume gross invoice equals PO consumption.
- **Currency and tolerance:** follow existing DICE rules; define whether tolerance applies to entitlement.
- **Approver changes role later:** approval stays valid; audit retains the actor.

22. Migration and Backward Compatibility

- Default all existing POs to Standard; no data conversion.
- Feature flag per entity and per PO category to control rollout.
- If existing partial invoicing lacks allocation records, backfill consumption from posted invoices only for POs explicitly switched to Progressive with Procurement sign-off.
- The Standard-flow regression suite must pass before every release.

23. Implementation Plan

Phase	Build	Exit condition
0. Fit-gap discovery	Inspect PO, GRN/SRN, invoice, matching, status and allocation model; list reusable services	Engineering documents what exists and what is missing
1. Shared entitlement control	Receipt-level allocations/balances, transactional reservation, audit, regression tests	Duplicate/excess allocation prevented without changing Standard POs
2. Progressive receipt billing	PO config, invoice selector, visibility, missing allocation behaviour	Receipt-based use cases pass
3. Milestone billing	Schedule builder, validation, evidence, approvals, invoice allocation, amendment history	Milestone use cases pass including SRN gate
4. Certified work progress (optional)	Claim model, certification workflow, cumulative calculation	Use cases pass, or explicitly out of scope
5. Hardening	Tax/adjustments, security, performance, monitoring, reporting	QA, AP, Procurement and Product sign-off

Rollout: internal pilot with 2 to 3 POs per method, then limited release for selected categories and suppliers, then general availability. Each stage has a go/no-go review on allocation integrity and AP feedback.

24. Use Cases and Acceptance Criteria

ID	Scenario	Expected result
UC-01	Standard PO regression	Invoice processing unchanged; no billing gate applied
UC-02	Partial accepted GRN (30 of 100 units)	Allocation of 30 succeeds; remaining eligible zero
UC-03	Excess receipt allocation (20 remain, 25 requested)	Blocked; remaining quantity displayed

ID	Scenario	Expected result
UC-04	Multiple invoices / multiple GRNs	Cumulative allocation never exceeds receipt or PO entitlement
UC-05	Pending/rejected GRN/SRN	Not usable for invoicing
UC-06	Concurrent allocation	Combined committed amount never exceeds remaining entitlement
UC-07	Certified progress (30,000 to 50,000 cumulative)	Eligible claim is AED 20,000 after certification
UC-08	Schedule totals AED 120,000 on AED 100,000 base	Cannot be finalised
UC-09	Approved AED 20,000 milestone, gates passed	Invoice of AED 20,000 succeeds; zero balance
UC-10	Second full-value invoice on fully allocated milestone	Blocked
UC-11	Service milestone approved, SRN pending	Remains ineligible
UC-12	Rejected/cancelled in-flight invoice	Reservation released; posted invoices use reversal/credit note
UC-13	PO amendment	Prior versions and allocations preserved; future entitlement uses approved version
UC-14	Duplicate supplier invoice	Existing duplicate detection still runs
UC-15	Credit/debit note and advance	Existing rules apply; receipt quantity not auto-restored by a financial credit
UC-16	Tax/currency	Existing DICE rules; entitlement is net, not gross payable
UC-17	Authorisation and audit	Only authorised users configure/approve; all events auditable
UC-18	Partial milestone invoice (if enabled)	AED 10,000 of 20,000 allocated; remaining AED 10,000
UC-19	Submitter attempts to approve own milestone	Blocked by segregation of duties

ID	Scenario	Expected result
UC-20	Double-click on invoice submit	One allocation created (idempotency)
UC-21	Receipt reduced after partial invoicing	Future entitlement reduced; posted invoices preserved and flagged
UC-22	Milestone rejected then resubmitted	History retained; eligible only after new approval

25. Test Strategy

- **Unit:** eligibility and balance calculations, schedule validation, cumulative claim maths.
- **Integration:** PO → receipt → allocation → invoice → matching → AP for each billing method.
- **Concurrency:** parallel submissions on the same balance; retry storms; lock timeouts.
- **Regression:** full Standard-flow suite on each release.
- **Negative:** overbilling, unapproved milestone, wrong supplier, cancelled PO, closed period.
- **UAT:** Procurement, AP and Project users run scripted scenarios on realistic POs.
- **Data integrity:** nightly reconciliation of allocations vs invoice totals vs PO limits.

26. Risks and Mitigations

Risk	Impact	Mitigation
Existing partial invoicing does not track consumption reliably	Overbilling	Phase 0 fit-gap; Phase 1 hardening before new features
Milestone and SRN confusion	Wrong eligibility	Explicit gate configuration and UI labelling
Concurrency bugs	Duplicate payments	Transactional reservation, idempotency, concurrency test suite
Scope creep into retention/advance logic	Delay	Treat as separate decisions; v1 documents behaviour only
Schedule amendments after billing	Audit exposure	Mandatory versioning and approval

Risk	Impact	Mitigation
Slow approvers	Delayed supplier payments	Notifications, ageing reports, delegation rules

27. Assumptions and Dependencies

- Existing GRN/SRN acceptance statuses are authoritative for receipt evidence.
- The PO amendment workflow can be reused for schedule changes.
- Notification and attachment services are reusable.
- Finance owns the rules for advances, retention and tolerance; this PRD depends on their confirmation.

28. Decisions Required Before Development

1. Is certified work-progress billing needed in the first release, or is accepted GRN/SRN sufficient?
2. Can the current invoice model allocate partial quantities from one GRN/SRN to multiple invoices?
3. Is the first milestone release one invoice per milestone, or partial milestone invoicing?
4. Which service categories require an accepted SRN before milestone invoicing?
5. How do current rules treat advances, retention, credit/debit notes, tolerance and tax in PO remaining balance?
6. Can billing method or schedule change after receipts/invoices exist, and what approval path applies?
7. At which invoice status does a reservation become a commitment (submitted, approved or posted)?
8. Who may approve milestones: PO creator, project owner, or a configured approver by value threshold?
9. Are milestone dependencies (M2 requires M1) enforced or advisory?
10. Are advance and retention modelled as milestones or handled outside the schedule?

29. Definition of Done

- PO supports billing method and configuration; existing POs retain current behaviour.
- Receipt-based progressive billing reuses GRN/SRN and safely tracks partial allocations.

- Milestone schedule, evidence, approval, SRN gate and invoice allocation are implemented.
- Certified work progress is implemented only if in scope, with cumulative certification controls.
- All allocations are concurrency-safe, auditable and reversible according to invoice state.
- Invoice matching and AP workflow are reused; no parallel invoice engine exists.
- Critical use cases and Standard-flow regression tests pass.
- Product, Engineering, QA, Procurement and AP sign off on the confirmed decisions.

30. Document Control

Version	Date	Change
1.1	24 Oct 2026	Implementation-focused draft
1.2	24 Oct 2026	Added goals/metrics, roles, deeper progressive and milestone design, requirements, permissions, notifications, reporting, migration, test strategy, risks, expanded use cases and decisions